

OMI F-Column Gauge Ladder and Byte-Ring Closure

Status

Filed reference synthesis

Layer: notation / gauge table / OMI-Port / OMI-ISA witness alignment

Current implementation impact: documentation first; bounded fixture alignment second

1. The Hidden Control Gauge Line

The non-printing control plane is canonical:

```
0x1C  FS  File Separator
0x1D  GS  Group Separator
0x1E  RS  Record Separator
0x1F  US  Unit Separator
```

This is the hidden gauge-control line.

It is the tangential gauge operator of the system because it sits at the end of the monotonic pre-language control span:

```
0x00 ... 0x1F
```

The control plane reaches its terminal local unit at:

```
0x1F  US
```

Then readable notation begins at:

```
0x20  SP
```

So the first boundary is:

```
Pinch Point:
0x00 ... 0x1F
```

Gauge Control:
0x1C ... 0x1F

Readable Space:
0x20

Interpretation:

The invisible control plane establishes place.
Space opens readable separation.
US is the hidden unit separator before printable notation begins.

2. The F-Column Printable Branch

From the hidden `US` point, OMI can read a printable vertical branch through the ASCII F-column:

0x1F	US	hidden unit separator
0x2F	/	printable separator
0x3F	?	witness/query separator
0x4F	0	upper/norm Omicron
0x5F	_	carrier/floor separator
0x6F	o	lower/local omicron

This gives the branch:

US → / → ? → 0 → _ → o

The F-column is the printable continuation of the hidden unit separator.

It is not borrowed from URI syntax.

It is OMI-native notation.

Canonical compact alias:

omi---imo?0_o

Expanded surface:

```
omi---imo/?---?0---0_---_o---o
```

Readable interpretation:

```
omi---imo
  folded root relation

/
  printable separator opened from US

?
  witness/query boundary

0
  upper/norm Omicron

-
  carrier/floor separator

o
  lower/local omicron
```

Canonical rule:

```
#/ belongs to carrier compatibility.
?0_o belongs to OMI notation.
```

3. Gauge Table and Interpretive Space

The full low declaration surface is:

```
Gauge Table:
  0x00 ... 0x7F
```

The high-bit boundary opens at:

```
0x80
```

This is the interpretive space:

Interpretive Space:
0x80

The complementary surrogate annotation band is:

0x7F ... 0xFF

Interpretation:

0x00 ... 0x7F
local readable declaration and gauge table

0x80
high-bit interpretive threshold

0x7F ... 0xFF
complementary surrogate annotation band

This upper band still needs careful local/remote and private/public distinction.

A provisional reading is:

0x80 ... 0x8F
optional frame bindings

0x90 ... 0x9F
optional buffer bindings

These are not yet authority.

They are candidate annotation bands.

4. Metatron Alignment

Metatron owns the alternating witness pair:

0xAA55 / 0x55AA

Reading:

```
0xAA
    alternating high-first scribe mask

0x55
    alternating low-first complement mask

0xAA55
    balanced Metatron scribe witness

0x55AA
    inverse balanced Metatron witness
```

Canonical rule:

```
0xAA55 / 0x55AA belongs to Metatron.
Metatron scribes and projects.
Metatron does not validate.
```

5. Tetragrammatron Governor

Tetragrammatron owns fold/carry validation and the polyharmonic governor axis.

The fold witnesses are:

```
0x3C
    centered fold witness

0x78
    shifted/carry fold witness
```

Required fold laws:

```
0x78 == (0x3C << 1) masked to 8 bits
0x3C == (0x78 >> 1)
0x3C has 4 one bits and 4 zero bits
0x78 has 4 one bits and 4 zero bits
0x3C ^ 0x78 == 0x44
```

Canonical rule:

```
0x3C / 0x78 belongs to Tetragrammatron.  
Tetragrammatron validates and fold-carries.  
Tetragrammatron does not project or scribe.
```

6. Tetragrammatron Polyharmonic Governor Laws

The executable Tetragrammatron witness must also preserve the larger governor structure.

6.1 Governor Counts

```
3 clocks  
4 visible offsets  
5 governor modes
```

6.2 Three Clocks

```
Atomic Logic Clock  
Spectral Observer Clock  
Cosmic Orbit Clock
```

6.3 Four Visible Offsets

```
FS = 0x0001  
GS = 0x0010  
RS = 0x0100  
US = 0x1000
```

6.4 Offset Stepping

```
0x0001 << 4 == 0x0010  
0x0010 << 4 == 0x0100  
0x0100 << 4 == 0x1000
```

6.5 Five Governors

```
FACTS      p = -1  
RULES      p =  0  
CLOSURES   p =  1
```

```
COMBINATORS  p = 2
CONS         p = 3
```

6.6 Governor Order

```
exponents are consecutive from -1 through 3
RULES is the p=0 Genesis / equality pivot
FACTS and CONS are circular inverse endpoints
```

Canonical summary:

```
FACTS ground.
RULES permit.
CLOSURES seal.
COMBINATORS compose.
CONS embodies.
```

Or in existing OMI canon:

```
Rules declare.
Facts ground.
Closures seal.
Combinators compose.
CONS reduces.
```

7. Null Ring and Byte-Ring Closure

The Null Ring owns the byte closure:

```
0x00 / 0x20 / 0x7F / 0xFF
```

Required law:

```
(NULL . NULL) is the folded null seed.
```

Byte ring:

```
(0x00 . 0x20)
(0x20 . 0x7F)
(0x7F . 0xFF)
(0xFF . 0x00)
```

Dot relation is XOR:

```
0x00 ^ 0x20 = 0x20
0x20 ^ 0x7F = 0x5F
0x7F ^ 0xFF = 0x80
0xFF ^ 0x00 = 0xFF
```

Full witness closure:

```
0x20 ^ 0x5F ^ 0x80 ^ 0xFF = 0x00
```

Canonical rule:

```
0x00 / 0x20 / 0x7F / 0xFF belongs to the Null Ring.
The byte ring closes because the full witness XOR returns to 0x00.
```

8. OMI Citation Notation

The citation form may be read as:

```
|OMI---IMO>
```

This is a notation-citation form, not execution authority.

It begins with the folded carrier:

```
omi---imo
```

and unfolds through the F-column branch:

```
US / ? 0 _ o
```


Canonical alias:

```
omi---imo?0_o
```

Full reading:

```
omi---imo
  folded root relation

US/
  hidden unit separator becoming printable slash

?0_o
  F-column witness/norm/carrier/local surface
```

9. Place-Value Cascade

The place-value cascade may be read through high-nibble positions:

```
0x0000 0x1000 0x2000 0x3000
0x4000 0x5000 0x6000 0x7000
0x8000 0x9000 0xA000 0xB000
0xC000 0xD000 0xE000 0xF000
```

This is a 4×4 high-nibble address surface.

It is aligned with the Polybius / gauge table reading:

```
0 1 2 3
4 5 6 7
8 9 A B
C D E F
```

The lower control path and upper place-value path are related by projection, not by textual equality.

10. Authority Separation

The byte ladder assigns roles:

Null Ring:

0x00 / 0x20 / 0x7F / 0xFF

Tetragrammatron:

0x3C / 0x78

3 clocks / 4 offsets / 5 governors

fold-carry validation

Metatron:

0xAA55 / 0x55AA

balanced alternating scribe witness

Omicron:

F* ... F*

resolver horizon / gauge enclosure

Canonical boundary:

Omicron frames.

Tetragrammatron validates.

Metatron scribes.

Null Ring closes.

OMI-Port routes candidates.

Receipt accepts.

11. Implementation Guidance

NULL_RING_PORT.md

Must contain:

(NULL . NULL) folded null seed

byte-ring XOR witnesses

full witness closure to 0x00

proof boundary to Coq Null Ring closure

TETRAGRAMMATRON_WITNESS_PORT.md

Must contain:

```
0x3C centered fold witness
0x78 shifted/carry fold witness
3 clocks
4 visible offsets
5 governor modes
Miquel / incidence balance if used
Tetragrammatron validates/fold-carries
Tetragrammatron does not project or scribe
```

METATRON_WITNESS_PORT.md

Must contain:

```
0xAA55 balanced alternating witness
0x55AA inverse balanced alternating witness
Metatron scribes/projections only
Metatron does not validate
```

OMI-Port notation docs

Must state:

```
#/ is carrier compatibility syntax.
?O_o is canonical OMI F-column notation.
```

12. Final Canon

```
The hidden gauge line ends at US.
US opens printable separation as /.
The F-column continues as / ? O _ o.
omi---imo?O_o is the compact OMI F-column alias.

0x00 / 0x20 / 0x7F / 0xFF belongs to the Null Ring.
0x3C / 0x78 belongs to Tetragrammatron.
0xAA55 / 0x55AA belongs to Metatron.

Omicron frames.
Tetragrammatron validates.
Metatron scribes.
Null Ring closes.
```

OMI-Port routes candidates.
Receipt accepts.